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Research on Startup Process for Sensorless Control of PMSMs Based on I-F Method Combined With an Adaptive Compensator

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ABSTRACT Speed control of permanent magnet synchronous motors (PMSMs) without position sensors is preferable in some applications due to cost and mounting space concerns. For applications in which the operating speed is relatively stable, the dynamic performance and efficiency demands of PMSMs are not severe during the startup process. Under such circumstances, a sensorless controller based on back electromotive force (EMF) estimation combined with a simple startup strategy is a suitable solution. In this paper, an I - f (current-frequency) startup method and a sliding mode observer are adopted to develop a sensorless control system, and the startup process which employs the I - f method is analysed in detail. Based on the analysis, this paper proposes a simple and robust startup strategy with an adaptive compensator to realize the smooth transition from I - f control to EMF-based control. The proposed strategy allows quick startup from standstill under different load conditions without initial position estimation. Furthermore, the strategy does not introduce any additional control coefficients, and none of the original control coefficients need to be adjusted. The strategy is suitable for both surface and interior PMSMs, and a test bench based on an interior PMSM is built to validate its properties. The experimental results show that the motor can smoothly accelerate from standstill to its rated speed in less than 2 s in both loaded and non-loaded situations, and the transition between the two sensorless control methods does not cause any large speed or current fluctuations.

INDEX TERMS Adaptive compensator, back electromotive force, permanent magnet synchronous motor (PMSM), sensorless control, startup strategy.

I. INTRODUCTION

Permanent magnet synchronous motors (PMSMs) have been extensively adopted in industrial applications and electric vehicles in recent years due to their positive features such as high power density and high efficiency [1]. For applications such as pumps, fans, and compressors, dynamic performance is not as significant as drive motors of electric vehicles as a result of relatively stable operating load and speed. However, cost, operating environment, and mounting space are always badly restricted in these applications, resulting in the elimination of position sensors [2].

Various sensorless control strategies have been proposed to realize effective control of PMSMs without position

sensors over the last decades, and these strategies can be classified into two categories: signal injection methods and back electromotive force (EMF) estimation methods [3], [4]. Signal injection methods can detect the rotor position information from the measured current using filters and observers, after injecting a specific high-frequency voltage into the stator windings. These methods perform pretty well at standstill and low speeds [5], [6]. However, the injected signals have bad effects on the efficiency and stability of PMSMs [7]. For back EMF estimation methods, several kinds of observers, such as the sliding mode observer (SMO) [8], [9], the extended Kalman filter observer [10], [11], and the adaptive observer [12], are presented based on the mathematical model of PMSMs. These back EMF observers can give precise rotor position estimation at nominal and high speeds. However, their performance

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deteriorates drastically at low speeds because the EMF is too low to be detected [13]. To realize ideal sensorless control of PMSMs over full speed range, hybrid solutions which combine the two methods are proposed [14], [15].

Although the hybrid sensorless control methods have strengths in operating performance, researchers suffer from its complexity. For applications with stable operating load and speed, the dynamic performance and efficiency demands of PMSMs are not severe during the startup process, making the hybrid solutions too much of a good thing. Based on this consideration, a robust back EMF estimator with a simple startup strategy as a complement will be more suitable, as long as the startup strategy can accelerate the motor to a speed at which the back EMF is high enough.

The V/f (voltage/frequency) method, which is widely employed in induction motor control, is a conventional startup strategy. It is an open-loop control strategy without position or current feedback, and it is quite simple but unstable under load [16]. Another simple startup strategy is known as the I - f method, which has closed-loop current regulation and no position feedback. A ramp speed command is given, and it is integrated to obtain the reference rotor position. Then the current in the rotating reference frame is specified and maintained to a constant value during startup [17]. The motor can be smoothly accelerated from standstill without large torque ripples because of the effective control of the stator current. Moreover, none of the motor parameters are required in the I - f method [18].

It is worth pointing out that there always is a deviation between the reference rotor position and the real rotor position in the I - f method, which is positively related to the given current. For applications such as pumps, the reference current is usually set to a very large value in case for heavy loads during the startup process. Though the I - f method and the back EMF estimators are simple and robust enough, the transition strategy between the two methods is a key problem. However, proposals for this matter are relatively rare, and the research of this paper is mainly inspired by the work presented in [19] and [20]. In [19], a first-order lag compensator was employed to achieve the smooth transition from I - f control to EMF-based control during the accelerating process. The effect depends heavily on the compensator parameters, which makes it difficult to cope with different load situations. In [20], the transition was realized by specifying the gradient of the deviating angle according to the reference current, which decreases at a computed rate after the motor reaches the target speed. As the transition procedure is finished after acceleration, the startup process will last for a long time. Moreover, torque ripples will significantly increase if the I - f method is executed at high speed, which is illustrated in detail later.

In this paper, a quick, robust and simple startup strategy is introduced to achieve smooth acceleration of PMSMs from standstill to high speed. It reduces the startup time considerably by taking two different ramps for the speed command during the startup process. Additionally, an adaptive

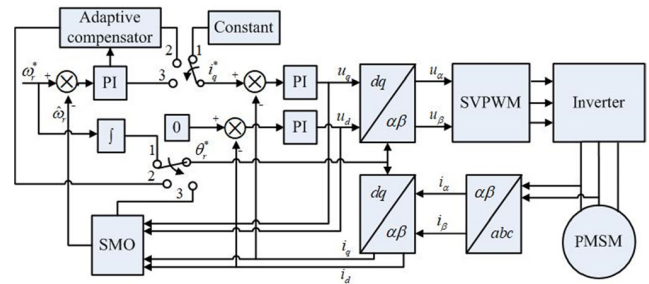


FIGURE 1. Structure of the proposed sensorless control system with a startup strategy consisting of three procedures.

compensator of deviating angle is designed based on the speed proportional-integral (PI) regulator to allow smooth transition between the I - f method and the back EMF estimation method. Compared with previous studies, this paper provides more detailed analysis of the I - f startup method, which not only proves its effectiveness without initial position estimation but also analyses its major drawbacks. These drawbacks will cause problems for previous methods but can be avoided with the help of the designed adaptive compensator. As a result, the startup strategy proposed in this paper can realize quicker startup without introducing any additional control coefficients. The proposed startup strategy is dedicated to sensorless control systems for applications such as pumps, fans and compressors, where operating load and speed are relatively stable and steady operation at low speed is not desired.

The paper is organized as follows. The structure of the sensorless control system is introduced in Section II. Then the startup strategy is described and analysed in Section III. The performance of the startup strategy is verified by experimental results in Section IV. Finally, the key features of the startup strategy are summarized in Section V.

II. SENSORLESS CONTROL SYSTEM

The structure of the proposed sensorless control system, which can be divided into three consecutive procedures, is shown in Fig. 1. First, the I - f method is employed to start the motor from standstill at low acceleration, with both switches connected to terminal 1 in Fig. 1. Then, the switches are switched to terminal 2, which means that the speed control loop is closed with the estimated speed from the back EMF estimator, and the motor can accelerate rapidly to operating speed because of the good dynamic response of PI regulators. During this procedure, the output of the back EMF estimator and speed PI regulator are further processed by an adaptive compensator so that the reference of rotor position and quadrature current component can gradually approach their ideal values. Finally, when the deviating angle between the reference position and the estimated position is small enough, the traditional field-oriented control (FOC) algorithm is employed with switches switched to terminal 3. For simplicity, these three procedures are called I - f startup, method transition and traditional FOC [21].

Among all the back EMF estimation methods, the SMO method is extensively studied and employed because of its robustness and simplicity. It can be executed in either stationary reference frame or rotating reference frame [22], [23], and the latter one is employed in this paper. The model of PMSM in the rotating reference frame can be written as follows [24]:

$$\begin{cases} \frac{di_d}{dt} = -\frac{R}{L_d}i_d + \frac{\omega_r L_q i_q}{L_d} + \frac{u_d}{L_d} \\ \frac{di_q}{dt} = -\frac{R}{L_q}i_q - \frac{\omega_r L_d i_d}{L_q} + \frac{u_q}{L_q} - \frac{\omega_r \psi_f}{L_q} \end{cases} \quad (1)$$

where u_d , u_q , i_d , i_q , L_d , and L_q are the stator voltage components, stator current components and inductances along the direct axis (d-axis) and the quadrature axis (q-axis), ω_r is electrical speed of the motor, R is the phase winding resistance, and ψ_f is the permanent magnet flux linkage. The structure of the SMO based on the model in (1) can be expressed as follows:

$$\begin{cases} \frac{d\hat{i}_d}{dt} = -\frac{R}{L_d}\hat{i}_d + \frac{\hat{\omega}_r L_q \hat{i}_q}{L_d} + \frac{u_d}{L_d} + ks_d + \varepsilon F(s_d) \\ \frac{d\hat{i}_q}{dt} = -\frac{R}{L_q}\hat{i}_q - \frac{\hat{\omega}_r L_d \hat{i}_d}{L_q} + \frac{u_q}{L_q} - \frac{\hat{\omega}_r \psi_f}{L_q} + ks_q + \varepsilon F(s_q) \end{cases} \quad (2)$$

where $\hat{i}_d$, $\hat{i}_q$, and $\hat{\omega}_r$ are the estimated quantities of i_d , i_q , ω_r , k and ε are gains of the observer, and s_d and s_q constitute the sliding surface S , which is defined as:

$$S = [s_d, s_q]^T = [i_d - \hat{i}_d, i_q - \hat{i}_q]^T \quad (3)$$

$F(x)$ is a sigmoid function described in (4), which can reduce the chattering phenomenon [25].

$$F(x) = \frac{2}{(1 + e^{-\alpha x})} - 1, \quad \alpha > 0 \quad (4)$$

where α is an adjustable parameter.

With proper values of k and ε , the estimation errors can converge to $S = 0$. The expressions of $\hat{\omega}_r$ and the estimated rotor position $\hat{\theta}_r$ are deduced as (5)-(7) to guarantee the stability of the SMO using a PI mechanism, which is analysed and demonstrated in [26].

$$\Delta \hat{\omega}_r = \frac{L_q}{L_d} \hat{i}_q s_d - \frac{L_d}{L_q} \hat{i}_d s_q - \frac{\psi_f}{L_q} s_q \quad (5)$$

$$\hat{\omega}_r = K_p \Delta \hat{\omega}_r + K_i \int \Delta \hat{\omega}_r dt \quad (6)$$

$$\hat{\theta}_r = \int \hat{\omega}_r dt \quad (7)$$

where K_p and K_i are the gains of the PI mechanism.

The SMO can precisely estimate the rotor speed and position at nominal and high speeds even taking the variation of motor parameters and the nonlinearity of inverter into consideration [27]. The startup strategy, which includes the procedures of I - f startup and method transition, will accelerate the motor to the effective speed interval of the SMO, which is discussed in Section III.

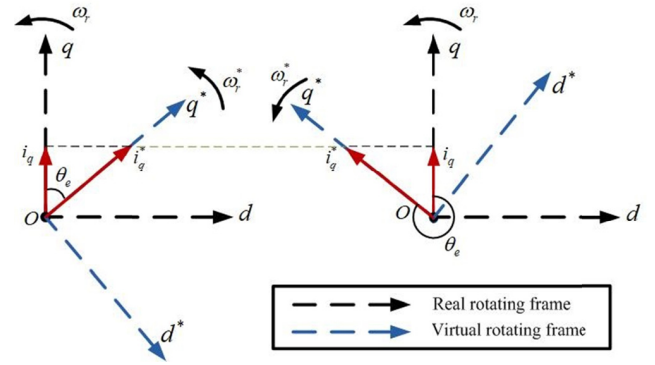


FIGURE 2. Diagram of the virtual rotating frame and the real rotating frame. The two situations generate equal torque, but only the left one can achieve dynamic balance.

III. STARTUP STRATEGY

A. ANALYSIS OF I-F STARTUP

The I - f startup method is employed here to start the motor from standstill and accelerate it to the effective speed interval of the SMO. During this procedure, the reference q-axis current is maintained constant and the reference d-axis current is set to zero. The reference rotor position is obtained by integrating the ramp speed command, as shown in (8)-(9).

$$\omega_r^* = K_\omega^* t \quad (8)$$

$$\theta_r^* = \int \omega_r^* dt \quad (9)$$

where ω_r^* is the speed command, θ_r^* is the reference position and K_ω^* is a constant. To analyse the dynamic characteristics of the startup procedure, a virtual synchronous rotating reference frame based on θ_r^* is established, as demonstrated in Fig. 2 along with the real rotating frame, and the deviating angle between the real frame and the virtual one is defined as θ_e .

For PMSMs, the motor torque can be expressed as [28]:

$$T = 1.5n_p i_q [\psi_f - (L_q - L_d)i_d] \quad (10)$$

where n_p is the number of pole pairs. In consideration of the deviating angle between the real frame and the virtual frame, i.e. θ_e , the stator current can be expressed as:

$$\begin{cases} i_q = i_q^* \cos \theta_e - i_d^* \sin \theta_e \\ i_d = i_q^* \sin \theta_e + i_d^* \cos \theta_e \end{cases} \quad (11)$$

where i_d^* and i_q^* are the reference current along the virtual d-axis and q-axis. Substituting (11) into (10), the motor torque can be rewritten as (12) because the reference d-axis current is set to zero.

$$T = 1.5n_p i_q^* \cos \theta_e [\psi_f - (L_q - L_d)i_q^* \sin \theta_e] \approx 1.5n_p i_q^* \cos \theta_e \psi_f \quad (12)$$

The term $(L_q - L_d)i_q^* \sin \theta_e$ in (12) can be omitted compared to the permanent magnet flux linkage. Then the state equation

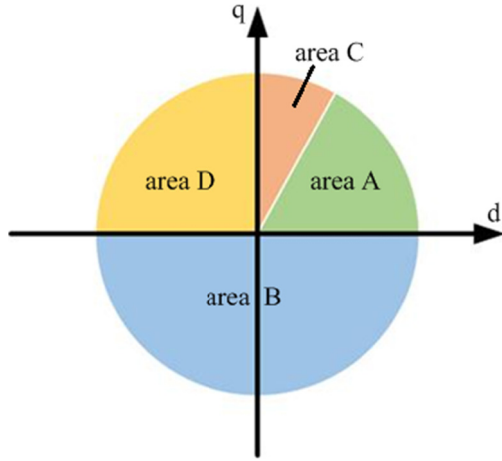


FIGURE 3. Area of the q-axis of the virtual frame in the real rotating frame before startup.

of the startup procedure can be established as follows:

$$\begin{cases} \frac{d\theta_e}{dt} = \omega_r - \omega_r^* \\ \frac{d^2\theta_e}{dt^2} = \frac{1.5n_p i_q^* \cos \theta_e \psi_f - T_L}{J} - K_\omega^* \end{cases} \quad (13)$$

where T_L is the load torque and J is the inertia of the motor.

The dynamic balance of the system can be achieved only if both the first derivative and the second derivative of θ_e are maintained at zero or fluctuates around zero. The limiting value of K_ω^* is then obtained in (14) because the second derivative of θ_e cannot be constantly positive under any load conditions.

$$K_\omega^* < \frac{1.5n_p i_q^* \psi_f - T_{L,\max}}{J} \quad (14)$$

where $T_{L,\max}$ is the maximum load torque during startup. The limitation of K_ω^* will then extend the time of the startup procedure, and a relatively small value of K_ω^* will reduce the speed fluctuation during startup.

In addition to the limitation of K_ω^* , there is also a limitation for θ_e to achieve the dynamic balance of PMSMs. First, $\cos \theta_e$ must be positive to balance the load torque. Then, $\cos \theta_e$ must be positively related to the load torque so that the motor torque can correctly respond to load changes. To be specific, there are two balance points for a constant load, as exhibited in Fig. 2. However, as the load torque increases, the actual acceleration of motor will decrease and θ_e will become smaller. Under this circumstance, only when $0 < \theta_e < \pi/2$, $\cos \theta_e$ will increase as θ_e decreases, and the motor torque will increase to achieve a new balance. A similar situation occurs when the load torque decreases.

Fortunately, θ_e can be automatically adjusted to its effective interval with a relatively small K_ω^* when the motor is started from standstill. When the q-axis of the virtual frame lies in area A in Fig. 3, the initial motor torque will be rather small and ω_r^* will be larger than ω_r . Then the motor torque will

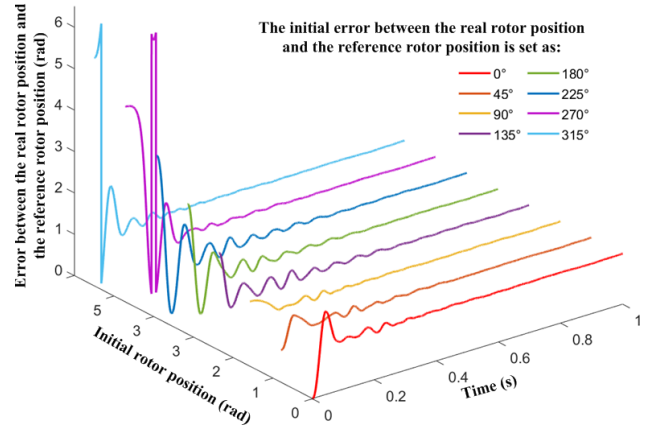


FIGURE 4. The error between the real rotor position and the reference rotor positions during the startup process with eight different initial rotor position. The initial value of the reference rotor position is set to zero.

increase gradually as θ_e decreases. The dynamic balance will be established soon as illustrated above. If the q-axis of the virtual frame lies in area B at the beginning, the initial torque will be negative, and the motor will rotate in the reverse direction. With a positive ω_r^* and a negative ω_r , θ_e will decrease until the virtual q-axis enters area A. For the condition in area C, the initial motor torque will be very large, and the initial acceleration will be too big to establish the balance right away. The virtual q-axis will firstly get into area A or area B, as ω_r is much larger than ω_r^* , and the balance will be realized after some dithering. Finally, in regard to area D, any fluctuation of the load torque will cause opposite changes of the motor torque, forming a positive feedback mechanism. If the motor torque decreases continuously, the virtual q-axis will soon enter area B and then area A. If the motor torque increases at the startup moment, the acceleration of θ_e will be kept positive until the motor torque decreases gradually to achieve the balance with the virtual q-axis entering area A. Another special working condition needed to be mentioned here is when the initial deviating angle is about 180 degree. Under this circumstance, the initial motor torque is close to zero and the real rotor position will not change. With a positive ω_r^* , the virtual q-axis will enter area B, and the motor state will be similar to that when the initial virtual q-axis lies in area B.

As K_ω^* is limited to a small value, neither ω_r^* nor ω_r will be too high to realize the relative position changes between the real frame and the virtual frame mentioned above. With a rather low motor speed, changes of the relative position will be accomplished within the first several motor rotations, where a few vibrations are tolerable. Simulation results of the motor startup process from standstill with eight different initial rotor positions are shown in Fig.4. The reference position is obtained by integrating the ramp speed command from zero. Among the eight curves in Fig.4, the green curve shows the motor state when the initial deviating angle is about 180 degree, which coincides with the theoretical analysis above. Actually, all of the simulation results are

entirely consistent with the analysis of different startup circumstances. It can be deduced that the motor has been started successfully without initial position estimation. It is worth noting that the analysis above is valid only for the process of starting the motor from standstill. If the motor has been accelerated to a relatively high speed, θ_e must be kept in its effective interval, or the motor will immediately fall out of step. To keep θ_e in its effective interval, i_q^* must be big enough to cover all possible loads. The reasonable value range of i_q^* can be derived from (14) as follows:

$$i_q^* > \frac{JK_\omega^* + T_{L,\max}}{1.5n_p\psi_f} \quad (15)$$

It can be deduced from (15) that motor torque generated by i_q^* not only needs to balance the load torque but also provide the acceleration during the startup process. If the motor load keeps unchanged after the startup process, i_q^* must be larger than the steady-state q-axis current. Generally, i_q^* is set to the rated current of the motor, as the startup load would be lower than the rated torque of the motor for the applications discussed in this paper.

As i_q^* is set as a positive constant, the motor can only rotate in one direction. In other words, the I - f startup method cannot response to reverse instructions, e.g. -1000 rpm/s. Fortunately, most of the applications discussed in the paper can only work effectively with the motor rotating in one direction. Some kinds of compressors can work in different rotating directions, but there is no need to change their rotating directions. In practical applications, reverse instructions can only appear in exceptional circumstances, e.g. when the three-phase wires are wrongly coupled. Under this circumstance, we can change the rotating direction by changing the phase sequence using software.

Another problem to note is that the performance of the I - f startup method deteriorates with high motor speed and light loads. In the digital control system, the rotor position is refreshed with high frequency, normally 4-10 kHz. In other words, the reference rotor position used in the digital controller is unchanged during every refresh cycle. However, the real rotor position is changing continuously, forming an additional deviation of the rotor position. And as the motor speed increases, the rotated angle of the motor during every refresh cycle will become larger. Assuming that the rotor position is exactly right to keep the motor operating steadily at the inception of a refresh cycle, the output torque of the motor at this moment can be derived according to (12):

$$T_{out,ini} = 1.5n_p i_q^* \cos \theta_e \psi_f \quad (16)$$

And the output torque at the end of the refresh cycle can be expressed as:

$$T_{out,end} = 1.5n_p i_q^* \cos(\theta_e + \frac{n_p \omega_r}{f}) \psi_f \quad (17)$$

where f is the refresh frequency of reference rotor position. Then the variation of the motor torque during every refresh

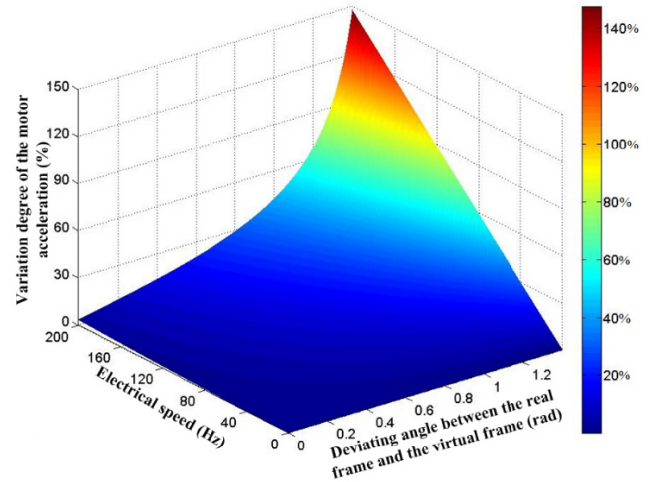


FIGURE 5. Degree of the motor acceleration regarding the electrical speed and the deviating angle between the two frames.

cycle can be expressed as follows:

$$\Delta T = 1.5n_p i_q^* \psi_f [\cos \theta_e - \cos(\theta_e + \frac{n_p \omega_r}{f})] \quad (18)$$

It is rational to treat the load torque as a constant value in one refresh cycle, and the relative variation of motor acceleration can be derived as:

$$\delta_a = \frac{\Delta T}{T_{out,ini} - T_L} > \frac{\Delta T}{T_{out,ini}} = 1 - \frac{\cos(\theta_e + \frac{n_p \omega_r}{f})}{\cos \theta_e} \quad (19)$$

A new variable is defined in (20) to measure the variation degree of acceleration, which also indicates the stability of motor operation during the I - f startup procedure.

$$\delta'_a = \left| 1 - \frac{\cos(\theta_e + \frac{n_p \omega_r}{f})}{\cos \theta_e} \right|, \quad 0 < \theta_e < \frac{\pi}{2} \quad (20)$$

If the load is heavy enough, θ_e will be relatively small and δ'_a will be kept within a tolerable range. However, δ'_a will become a considerable value under light load and non-load conditions, as θ_e is close to $\pi/2$. For example, suppose that the refresh frequency is 5 kHz, and the motor with four pole pairs has been accelerated to 1000 rpm. If θ_e is $2\pi/5$ at the moment, δ'_a would be 26.1%, and if θ_e is $4\pi/9$, δ'_a would be 47.8%. The relationships among δ'_a , ω_r and θ_e are further analysed in Fig. 5, which indicates that torque and acceleration ripples will significantly increase if the I - f method is employed at high speed under light loads. Simulation results of the startup process from standstill under no load also supports this opinion. Speed fluctuations can be observed both at the beginning stage and the high speed stage in Fig. 6. The initial fluctuations decrease rapidly and disappear in a short time, while as the speed increases, there is a tendency for the fluctuations to increase. It has to be pointed out that similar issues also occur when traditional FOC and position sensors are employed. However, the deviating angle between the real frame and the reference frame, denoted as θ_e ,

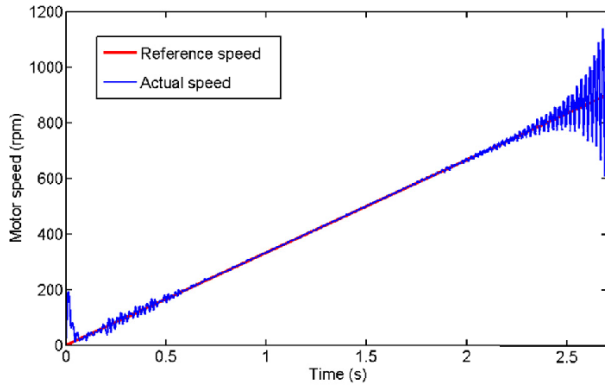


FIGURE 6. Results of the motor speed during the startup process under no load using the I - f method.

is approximately zero, making δ'_a ignorable in traditional FOC.

Based on the analysis above, there are two main drawbacks when the I - f startup method is employed to accelerate the motor to its operating speed. First, the limitation of K_ω^* will extend the startup time, and fast startup is in contradiction with the stability of the startup process from standstill. The other problem is that significant fluctuations will occur as the motor speed rises under light load. A simple and robust method transition strategy is designed below to address these problems and achieve quick and reliable startup of the motor.

B. METHOD TRANSITION STRATEGY

As illustrated above, a smaller K_ω^* benefits the stability of the I - f startup procedure but extends the startup time. It is important to note that SMO can be quite effective when the electrical angular frequency reaches 10-15 Hz, usually 10-20% of the rated frequency. In other words, it is rational to switch into SMO sensorless control and realize a closed speed loop with the motor accelerated to about 10-20% of its operating speed. By establishing the closed speed loop ahead of time, a much bigger K_ω^* can be applied to reduce the startup time. Furthermore, the I - f startup would not be employed at high speed, which means that both drawbacks of the I - f startup method in the previous discussion can be avoided. To design a suitable reference speed curve, the limitation of K_ω^* given in (14) is mandatory during the I - f startup procedure. Besides, the requirements of practical applications must be taken into consideration. For example, the startup process of oil pumps should be finished within 3 s when used in electric hydraulic power steering system. After accounting for the factors above, it is recommended to set K_ω^* as small as possible to make the startup process more reliable. The reference speed curve can be found in Fig. 7, with K_ω^* changed from 100 rpm/s to 500 rpm/s when the electrical angular frequency reaches 10 Hz.

When K_ω^* changes, the estimated rotor speed and position are adopted from SMO. However, as i_q^* is set to a large value, there is always a deviation between the reference rotor position of the I - f method and the estimated one of SMO.

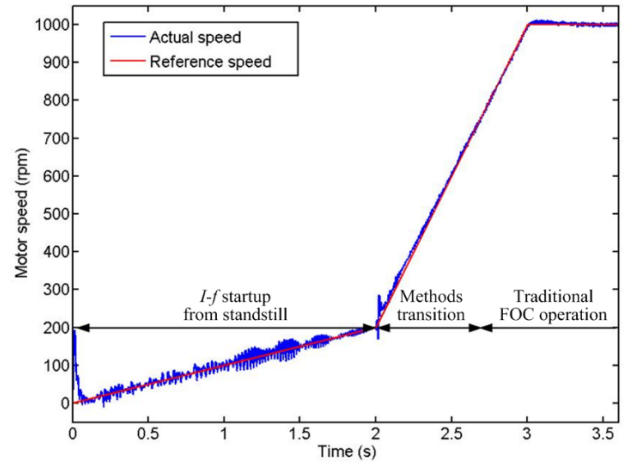


FIGURE 7. Reference speed and the motor speed during the startup process under no load using the proposed startup strategy.

The deviation will be much bigger if the motor operates under light load or non-load conditions. With the transition executed suddenly, the speed, torque and current of motor will fluctuate drastically due to the mismatch of the current amplitudes. Therefore, a reliable and robust method transition strategy is needed here to allow the rotor position and q-axis current to gradually approach ideal values. More specifically, the goal of the method transition strategy is to smoothly reduce θ_e to zero and the reference q-axis current to its ideal value.

As a complement to the speed PI regulator, an adaptive compensator of θ_e is designed here to realize the transition from the I - f method to the SMO method. In a classical speed PI regulator, the reference q-axis current is always regarded as the output. After each computation cycle, the reference q-axis current is refreshed based on the errors between the feedback motor speed and the reference motor speed. Then it is regarded as the input of the current PI regulator. It must be emphasized here that the output of the speed PI regulator should be the reference q-axis current in the real rotating frame, denoted as i_q^r . However, as θ_e is not zero, the current PI regulator can only realize control of the q-axis current in the virtual frame, but not the q-axis current in the real frame. In a word, during the method transition procedure, i_q^r is the output of the speed PI regulator and i_q^* is the input of the current PI regulator. The relationship between i_q^* and i_q^r can be described as:

$$i_q^r = i_q^* \cos \theta_e \quad (21)$$

As SMO is effective enough in the method transition procedure, θ_e can be obtained by regarding the estimated rotor position as the real rotor position, which means i_q^* can be calculated easily. However, it is of no help to the transition process. The relationship between the variation of θ_e , i_q^* and i_q^r can be derived by taking the derivative of (21):

$$\frac{di_q^r}{dt} = \cos \theta_e \frac{di_q^*}{dt} - i_q^* \sin \theta_e \frac{d\theta_e}{dt} \quad (22)$$

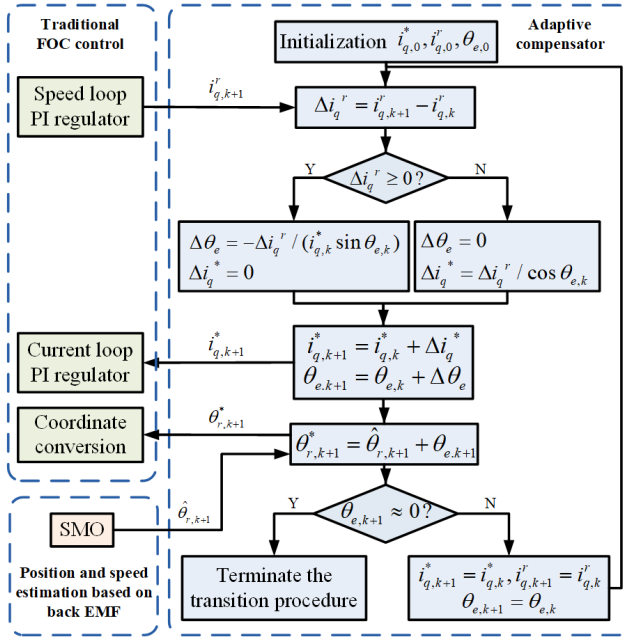


FIGURE 8. Algorithm flow diagram of the adaptive compensator in the discrete system.

Then (22) is discretized to obtain the following expression:

$$\Delta i_q^r = \Delta i_q^* \cos \theta_e - \Delta \theta_e i_q^* \sin \theta_e \quad (23)$$

where $\Delta \theta_e$, Δi_q^* and Δi_q^r are the variation of θ_e , i_q^* and i_q^r in every calculation cycle. Obviously, it is difficult to determine the variation of θ_e and i_q^* at the same time according to (23). However, it can be found from (21) that decreases in θ_e and i_q^* will lead to different changes in i_q^r . It is reasonable to allow only one variable to change under different conditions. When i_q^r increases, θ_e can be reduced while i_q^* remains unchanged. For the opposite case, i_q^* changes. With this adaptive compensation method, θ_e will be rapidly adjusted around zero, and the values of i_q^* and i_q^r will be almost equal. Thus, the purpose of the method transition procedure can be realized, and the control effect of the q-axis current will not be affected. The expression in (23) can be rewritten as follows:

$$\Delta i_q^r = \begin{cases} \Delta i_q^* \cos \theta_e, & \Delta i_q^r < 0 \\ -\Delta \theta_e i_q^* \sin \theta_e, & \Delta i_q^r \geq 0 \end{cases} \quad (24)$$

A clear description of the compensator is illustrated as follows, with Fig. 8 as a reference in the discrete system:

Step 1: The refreshed value of i_q^r is obtained from the speed PI regulator, and its variation is calculated using the last two values of i_q^r .

Step 2: The sign of Δi_q^r is determined, and the variation of θ_e and i_q^* can be calculated according to (24).

Step 3: The value of i_q^* is refreshed and sent to the current PI regulator as the input.

Step 4: The value of θ_e is refreshed and added to the estimation rotor position of SMO, making it the reference position during the transition procedure.

Step 5: Determine whether the value of θ_e is small enough to terminate the transition procedure.

At the beginning of the transition procedure, the initial values of i_q^* and i_q^r are obtained by inheriting i_q^* in the I - f startup procedure, and the initial value of θ_e is calculated by (25).

$$\theta_{e,0} = \theta_{r,0}^* - \hat{\theta}_{r,0} \quad (25)$$

where $\theta_{r,0}^*$ and $\hat{\theta}_{r,0}$ are the values of θ_r^* and $\hat{\theta}_r$ at the beginning of the method transition procedure.

The compensator above can guarantee that both θ_e and i_q^* approach their ideal values gradually without affecting the dynamic performance of the PI regulator. The varying speeds of θ_e and i_q^* mainly depend on the coefficients of the PI regulators. Generally, these PI regulators are tuned based on the motor parameters to guarantee the response characteristic of the control system. The simulation results in Fig. 7 indicate that proper PI coefficients for the control system are right enough to accomplish the transition procedure without large speed fluctuations. That is to say, it is not necessary to add any additional control coefficients or adjust any original control coefficients for the transition procedure.

To summarize, the startup strategy consists of three procedures:

Procedure 1: I - f startup from standstill. A ramp speed command with low changing rate is generated as the reference motor speed. The reference q-axis current of the virtual frame is set to a constant value to start the motor with a closed current loop.

Procedure 2: Method transition. As long as the motor reaches the effective speed interval of SMO, the estimated speed of SMO is used to close the speed loop, and the adaptive compensator is applied to realize a smooth transition of the q-axis current and the rotor position.

Procedure 3: Traditional FOC operation. After the transition, the reference q-axis current is generated by the speed PI regulator. The estimated motor speed and position from SMO are applied in the control system.

IV. EXPERIMENTAL RESULTS

The purpose of this section is to verify the properties of the proposed startup strategy. For the experiments, a home-made three-phase two-level IGBT inverter switching at 6 kHz and a DC voltage source of 540 V are applied. The strategy is realized with a digital controller based on DSP TMS320F28035 with a clock speed of 60 MHz [29]. The motor is loaded by a hydraulic circuit, which consists of a hydraulic oil pump, a hydraulic valve, an oil reservoir and high-pressure oil pipes. Furthermore, an optical encoder is installed between the tested PMSM and the pump to validate the motor speed and position estimation. The experimental setup is shown in Fig. 9. The parameters of the tested PMSM are given in Table 1.

The motor is started from standstill and accelerated to its rated speed in the experiment. The hydraulic valve is used to tune the load. The reference acceleration, i.e. K_ω^* , is set

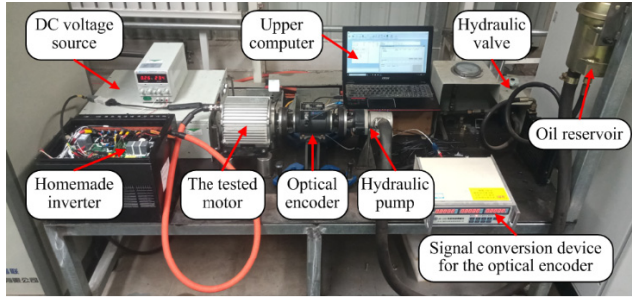


FIGURE 9. Experimental setup.

TABLE 1. Motor parameters.

Denotation	Symbol	Value
Rated power (kW)	P_R	1.5
Rated Current (A)	I_R	6
Rated speed (rpm)	U_R	1000
Rated load (N·m)	T_{LR}	15
Rated frequency (Hz)	F_R	66.7
Number of pole pairs	n_p	4
Stator resistance (Ω)	R	0.913
d-axis inductance (mH)	L_d	11.916
q-axis inductance (mH)	L_q	28.024
Flux linkage (Wb)	Ψ_f	0.248

to 120 rpm/s (10 Hz/s) during the I - f startup procedure and then switched to 1200 rpm/s (100 Hz/s) when the speed loop is closed at 180 rpm, which is 18% of the rated motor speed. The reference q-axis current is set to 6 A during the I - f startup procedure, which is the rated current of the tested motor. The performance of the startup strategy proposed in this paper is first demonstrated under no load. Then the motor is started under approximately 60% of the rated load to verify the startup performance under load. All the experimental results are shown in Fig. 10 and Fig. 11.

As the SMO in this paper is executed in the rotating reference frame, u_d , u_q , i_d and i_q in (1) and (3) need to be reconstructed using the estimated rotor position of SMO. For SMO in the stationary reference frame [30] and other sensorless control methods of PMSMs, the startup strategy is also valid as long as precise estimation of the rotor position can be obtained at nominal and high speeds.

Fig. 10 shows the operating status of the motor during the startup process under no load. The transition procedure starts at 1.400 s and ends at 1.934 s, while the reference speed reaches the rated motor speed at 1.911 s. An abrupt speed change at the initial startup moment can be seen in Fig. 10(a) due to the unknown initial rotor position. However, the motor speed stabilizes quickly because of the relatively small reference acceleration, as discussed in Section III-A. During the startup process, no large speed ripples can be observed, which means that the drawbacks of the I - f method are avoided and smooth transition between two sensorless control method is achieved. As can be found in Fig. 10(b), the q-axis current is

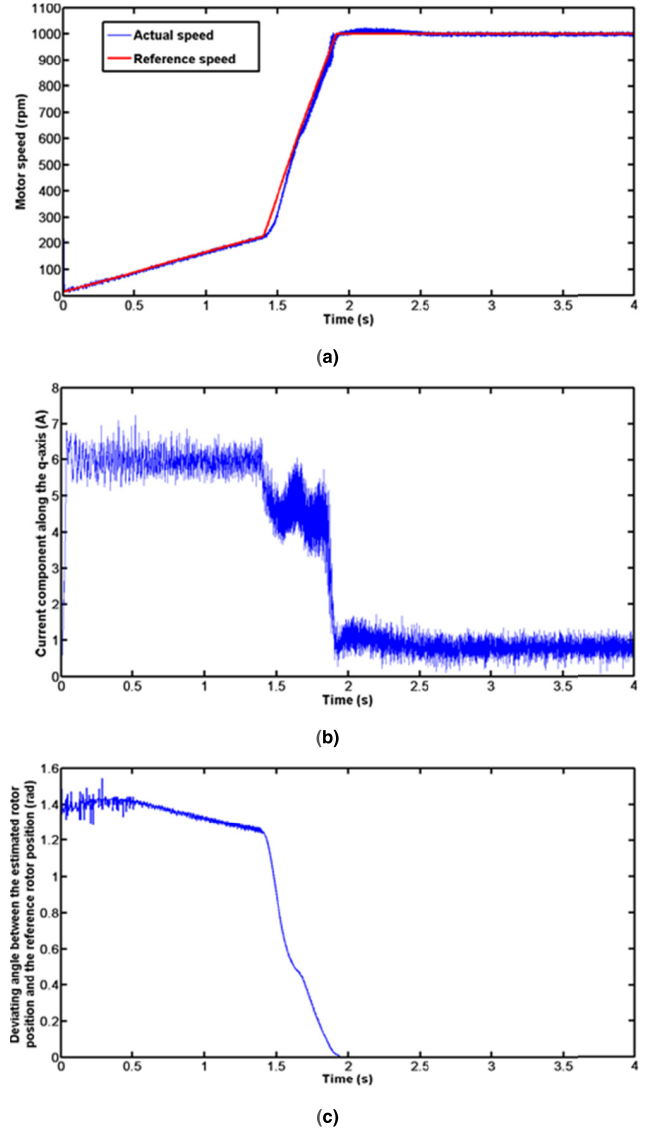


FIGURE 10. Status of the motor during the startup process under no load. (a) Comparison of reference speed and actual speed. (b) Current component along the q-axis. (c) Deviating angle between the reference rotor position and the estimated rotor position.

maintained at approximately 6 A before the transition procedure, and at approximately 0.9 A after the startup process. A few fluctuations of the q-axis current can be observed during the transition procedure because the change of the gradient of reference motor speed temporarily leads to larger torque requirement. Fig. 10(c) shows the value of θ_e , which fluctuates at low speed and stabilizes to approximately 1.2 rad when the motor speed rises to the effective interval of SMO. When the transition procedure begins, θ_e decreases rapidly. When θ_e is lower than the limiting value, which is 0.02 rad here, the transition procedure is terminated, and the estimated rotor position of SMO is employed as the reference rotor position.

Experimental results of the startup process under load are shown in Fig. 11. Compared with the startup process under no load, speed ripples during the transition procedure are minor,

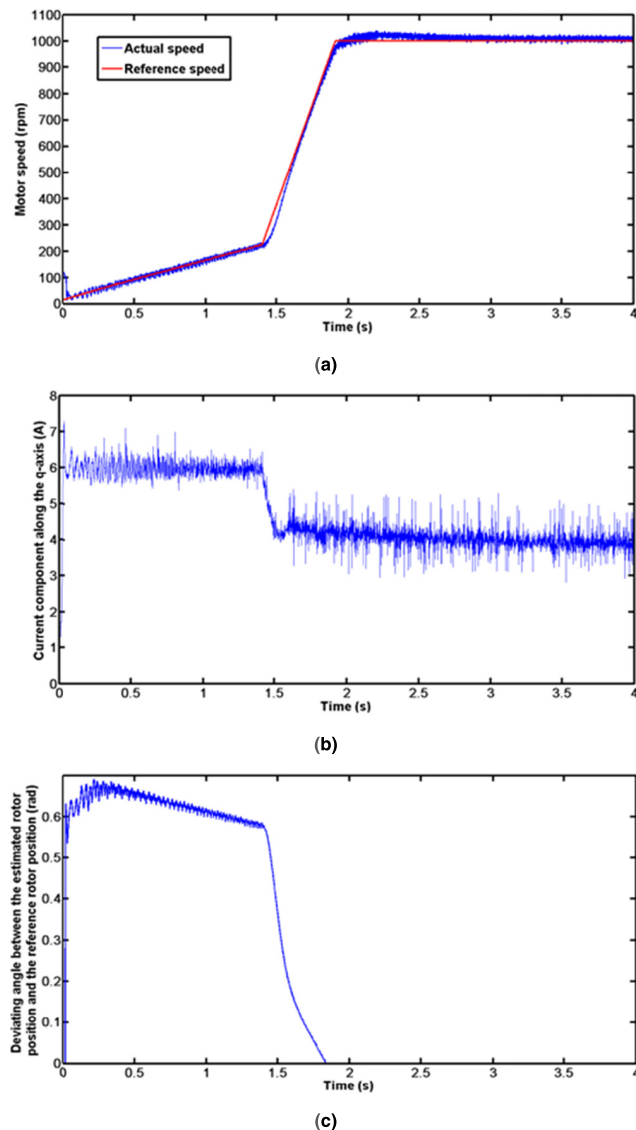


FIGURE 11. Status of the motor during the startup process under load. (a) Comparison of reference speed and actual speed. (b) Current component along the q-axis. (c) Deviating angle between the reference rotor position and the estimated rotor position.

but the speed change at the initial moment is more obvious in Fig. 11(a) due to different initial rotor positions. The q-axis current stabilizes to approximately 3.9 A in Fig. 11(b) with the same initial q-axis as in Fig. 10(b). It can be observed from Fig. 11(c) that θ_e is smaller than that of the startup process under no load, which is approximately 0.6 rad before the transition procedure because the required motor torque is much larger to balance the startup load. Furthermore, smaller θ_e results in a shorter transition procedure. As can be seen in Fig. 11(c), the transition procedure starts at 1.400 s and ends at 1.824 s, even before the reference speed reaches the rated speed.

It has to be pointed out that reverse rotation can be detected in the first 0.2 s of the startup process in both loaded and non-loaded situations, which is more obvious in simulation results in Fig. 4. As there is not severe load change during the

startup process, the reverse rotation does not have any effect on the running of motor. Actually, dynamic performance of the motor during the startup process are not demanding for applications discussed in this paper, so the reverse rotation in the startup process is acceptable.

In summary, it can be observed from the experimental results that the motor smoothly accelerates from standstill to its rated speed and the actual motor speed can completely track the reference values in both loaded and non-loaded situations. The startup process can be finished in quite a short time. Furthermore, the transition between two sensorless control methods does not cause any large speed or current fluctuations.

V. CONCLUSION

A novel startup strategy for sensorless control of PMSMs is proposed in this paper, which utilizes the *I-f* startup method and avoids its major drawbacks. In the proposed startup strategy, an adaptive compensator is designed to realize the smooth transition between different sensorless control methods. With this startup strategy, the motor can accelerate directly to its rated speed in a short time under different load conditions. Moreover, initial position estimation is not required, and no new parameters need to be added for this strategy, both of which greatly improve the robustness and simplicity of the control system. All the above conclusions have been verified by simulations and experiments. The strategy is suitable for industrial applications such as pumps, fans, and compressors, and it can be used to help researchers who focus on sensorless control methods based on back EMF estimation to realize simple and quick startup of PMSMs. Furthermore, the characteristics of the *I-f* method are analysed in detail in this paper, because the related research is rare and neglected. It is worthy of expectation that the *I-f* method can be studied in depth and applied extensively in the future.

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